nVipani

Set-Lake

Total Questions: 55 **Duration: 60 Minutes**

Instructions to candidates

1. Do not write anything on the question paper. Use a separate sheet of paper for any rough work.
2. Answers should be written on the answer sheet given.
3. Fill your full name, roll number, educational qualification with the branch, phone, email and address in the answer sheet.
4. Each Multiple Choice Question has one correct answer. Questions not attempted will be awarded zero marks. There is no negative or partial marking.
5. Blank paper sheets will be provided for rough work. These should be returned to the invigilator at the end of the exam along with the answer sheet and question paper.

Quantitative Ability

1. In an election between two candidates, 75% of the voters cast their votes, out of which 2% of the votes were declared invalid. A candidate got 9261 votes which were 75% of the total valid votes. Find the total number of voters enrolled in that election.
   A. 16600 B. 15600 C. 16800 D. 15800

2. Six bells commence ringing together and ring at intervals of 2, 4, 6, 8 10 and 12 seconds respectively. In 30 minutes, how many times do they ring together?
   A. 13 times B. 14 times C. 15 times D. 16 times

3. A batsman in his 16th innings makes a score of 170 runs and thereby increasing his average by 9 runs. Find his average after 16th innings.
   A. 27 B. 37 C. 25 D. 35

4. The current age of a father is 3 years more than three times the age of his son. Three years hence, the father's age will be 10 years more than twice the age of the son. What is the present age of the father?
   A. 33 B. 35 C. 43 D. 45

5. The number of digits used in numbering each page of a book of 150 pages is:
   A. 342 B. 348 C. 328 D. 322

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6. A is thrice as good a workman as B and therefore is able to finish a job in 60 days less than B. Working together, they can do it in:
A. 20 days B. 22 ½ days C. 25 days D. 30 days

7. A watch gains 5 seconds in 3 minutes and was set right at 8 AM. What time will it show at 10 PM on the same day?
A. 7 PM B. 10:31:7 PM C. 10:23:20 PM D. 10:23:20 AM

8. The dimensions of a certain machine are 48" x 30" x 52". If the size of the machine is increased proportionately until the sum of its dimensions equals 156", what will be the increase in the shortest side?
A. 6 B. 7 C. 26 D. 28

9. Three pipes A, B and C can fill a tank from empty to full in 30 minutes, 20 minutes and 10 minutes respectively. When the tank is empty, all the three pipes are opened. A, B and C discharge chemical solutions P, Q and R respectively. What is the proportion of solution R in the liquid in the tank after 3 minutes?
A. 5/11 B. 6/11 C. 7/11 D. 8/11

10. Gauri went to the stationers and bought things worth Rs. 25, out of which 30 paise went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax-free items?
A. Rs. 15 B. Rs. 15.70 C. Rs. 19.70 D. Rs. 20

11. In a 100 m race, P beats Q by 10 m and R by 13 m. If it is 180 m race, then Q will beat R by:
A. 6 B. 7 C. 4 D. 8

12. A vendor sells 50 percent of apples he had and throws away 20 percent of the remainder. Next day he sells 60 percent of the remainder and throws away the rest. What percent of his apples does the vendor throw away?
A. 20% B. 22% C. 24% D. 26%

13. A boatman goes 2 km against the current of the stream in 1 hour and goes 1 km with the current in 10 minutes. How long will it take to go 5 km in stationary water?
A. 40 min B. 1 hour C. 1 hr 15 min D. 1 hr 30 min

14. In a mixture of 60 litres, the ratio of milk and water 2:1. If this ratio is to be 1:2, then the quantity of water to be further added is:
A. 20 litres B. 30 litres C. 40 litres D. 60 litres

15. A circular wire of radius 42cm is cut and bent into the form of a rectangle whose sides are in the ratio of 6:5. The smaller side of the rectangle is:
A. 30cm B. 60cm C. 72cm D. 132cm

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Logical Ability

16. Charlie lives on the topmost floor. There is exactly one person between Ben and Annie. There are exactly two people between Feroz and David. Annie and Elly live on successive floors. Gina lives between Charlie and Feroz. Who lives on the seventh floor?
A. Gina B. Feroz C. Elly D. Charlie

17. Hypothetically, if Ooty is in the East of Darjeeling and Darjeeling is in the North of Varanasi, Surat is in the South of Varanasi, then in which direction of Ooty is Surat?
A. South-East B. South-West C. South D. North-East

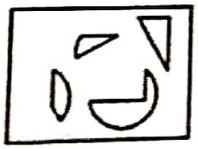
18. If A+B means A is the mother of B; A-B means a is the brother of B, A%B means A is the father of B and A x B means A is the sister of B, which of the following shows that P is the maternal uncle of Q?
A. Q - N +M x P B. P + S x N - Q C. P - M + N x Q D. Q - S % P

19. In a certain code, ARMY is written as ASOB. How is JUDGE written in that code?
A. KVEHF B. JVFJI C. ITCFD D. JWGKJ

20. Pointing to a photograph, a man said, "I have no brother, and that man's father is my father's son."Whose photograph was it?
A. His son B. His own C. His father D. His nephew

21. A is the husband of B. E is the daughter of C. A is the father of C. How is B related to E?
A. Mother B. Grandmother C. Aunt D. Cousin

22. Find out which of the figures (1), (2), (3) and (4) can be formed from the pieces given in figure (X).



(X)

(1) (2) (3) (4)

A. 1 B. 2 C. 3 D. 4

DIRECTIONS for questions 23 to 26: Answer the questions on the basis of the information given below.

Eight cities, A through H, are connected by a network of two-way roads as shown in the figure below:

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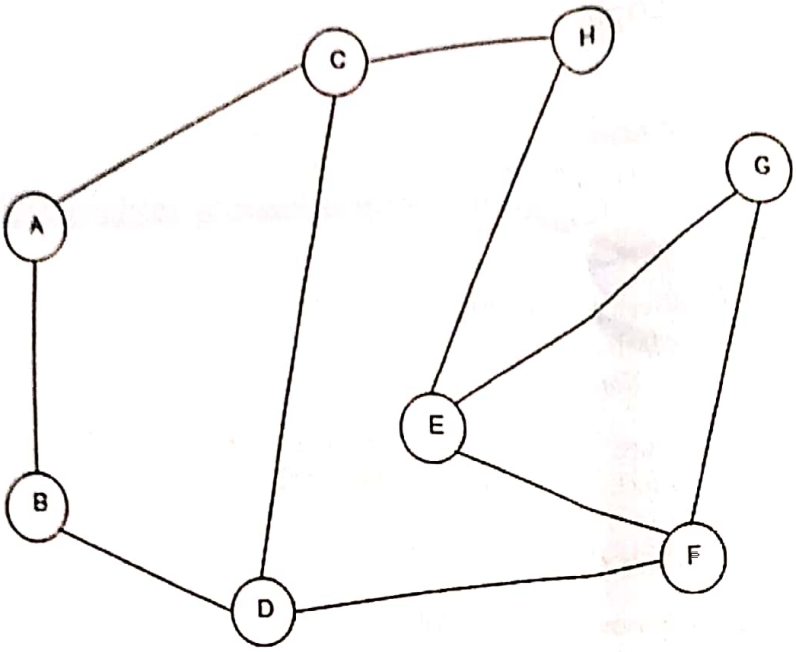

Kiran, a travelling salesman, has to travel from one of the eight cities to another, such that he visits each of the eight cities exactly once. The city from which he starts is referred to as the origin, while the city that he finally reaches is referred to as the destination.

23. If B must be the third city that Kiran visits (counting the origin as the first city), which of the following can be his destination?
A. G B. E C. F D. A

24. Which of the following pairs of cities cannot be the last two cities that he visits?
A. C,D B. E,F C. D,F D. More than one of the above

25. If the origin is E, how many of the other seven cities can be the destination?
A. 0 B. 1 C. 2 D. 3

26. If the origin and the destination should not be directly connected, how many of the eight cities can be the origin?
A. 4 B. 5 C. 3 D. 6

DIRECTIONS for questions 27 to 29: Answer the questions on the basis of the information given below.

Five roommates Randy, Sally, Terry, Uma and Vernon each do one housekeeping task mopping, sweeping, laundry, vacuuming, or dusting one day a week, Monday through Friday.

a. Vernon does not vacuum and does not do his task on Tuesday.
b. Sally does the dusting and does not do it on Monday or Friday.
c. Mopping is done on Thursday.
d. Terry does his task, which is not vacuuming, on Wednesday.
e. The laundry is done on Friday, and not by Uma.
f. Randy does his task on Monday.

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27. What task does Terry do on Wednesday?
A. Vacuuming B. Dusting C. Mopping D. Sweeping

28. What day does Uma do her task?
A. Monday B. Tuesday C. Wednesday D. Thursday

29. What day is the vacuuming done?
A. Friday B. Monday C. Tuesday D. Wednesday

30. **Read the following argument and answer the question below.**
One view of the impact of labour on valuation suggests that asking customers to assume production costs should result in reduced willingness to pay, once customers subtract the value of their labour from the overall cost of the product; examples, instead, suggest that when people imbue products with their own labour, their effort can increase their valuation. And while some labour is enjoyable (building a bear with one's nephew) and some labour allows for product customization (building a bear with one's alma mater's logo) – both of which might increase valuation – we suggest that labour alone can be sufficient to induce a greater liking for the fruits of one's labour: even constructing a standardized bureau, an arduous, solitary task, can lead people to overvalue their (often poorly constructed) creations. We call this phenomenon the "IKEA effect," in honour of the Swedish manufacturer whose products typically arrive with some assembly required.

Consider that an IKEA product involves a customer's labour whereas a non-IKEA product doesn't, all other parameters being similar. Which of the following, when proven, most ascertains the veracity of the author's claim?

A. Customers are more willing to pay for an IKEA product than to pay for a non-IKEA product.
B. Customers are willing to pay more for an IKEA product than for a non-IKEA product.
C. Customers are as willing to pay for an IKEA product as they are to pay for a non-IKEA product.
D. Customers are willing to pay more for an IKEA product than for a non-IKEA product, provided they enjoyed building the customised IKEA product.

Verbal Ability

Convert the following sentences from active voice to passive:

31. "They greet me cheerfully every morning."
A. Every morning I was greeted cheerfully.
B. I am greeted cheerfully by them every morning.
C. I am being greeted cheerfully by them every morning.
D. A cheerful greeting is done by them every morning to me.

32. You need to clean your shoes properly.
A. Your shoes are needed to clean properly
B. You are needed to clean your shoes properly
C. Your shoes need to be cleaned properly
D. Your shoes are needed by you to clean properly.

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Correct / Improvise the following sentences wherever required

33. If the room had been brighter, I would have been able to read for a while before bedtime.
A. If the room was brighter B. If the room is brighter
C. Had the room been brighter D. No Improvement

34. The performance of our players was rather worst than I had expected.
A. bad as I had expected B. worse than I had expected
C. worse than expectation D. worse than was expected
E. No correction required

35. The intruder stood quietly for few moments.
A. for few movements B. for the few moments
C. for moments D. for a few moments
E. No correction required.

36. Maria unnecessarily picked up a quarrel with Rani and left the party hurried.
A. has picked up B. picked on
C. picked D. picking up
E. No correction required.

37. This is one of the most important inventions of this century.
A. invention of this century B. invention of these century
C. invention of centuries D. invention of the centuries
E. No correction required

38. My uncle decided to take ..... and my sister to Disneyland.
A. I B. myself C. me D. Mine

39. I put ..... the light and slept.
A. up B. down C. in D. out

40. Select the word that is closest in meaning to stealing secretly in small quantities.
A. Pilferage B. Theft C. Defalcation D. Robbery

Technical Ability

41. The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the postorder traversal sequence of the same tree?
A. 10, 20, 15, 23, 25, 35, 42, 39, 30 B. 15, 10, 25, 23, 20, 42, 35, 39, 30
C. 15, 20, 10, 23, 25, 42, 35, 39, 30 D. 15, 10, 23, 25, 20, 35, 42, 39, 30

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42. What is the worst case time complexity of the Quick sort?

A. O(nlogn) B. O(n) C. O(n3) D. O(n2)

43. Assume that the following jobs are to be executed on a single processor system

Job-Id	CPU-BurstTime
p	4
q	1
r	8
s	1
t	2

The jobs are assumed to have arrived at time 0 and in the order p, q, r, s, t. Calculate the departure time (completion time) for job p if scheduling is round robin with time slice 1.

A. 4 B. 10 C. 11 D. 12

44. What is the output of below javascript code:

```
var d =3;
var e=""3"";
console.log(d == e);
console.log(d === e);
```

A.true false B. true true C. false false D. false true

45. What is the output of the below java code:

```
class Bank
{
    public void getInterestRate()
    {
        System.out.println(""5%"");
    }
}
public class SBI extends Bank
{
    public void getInterestRate()
    {
        System.out.println(""6%"");
    }
    public static void main(String[] args)
    {
        Bank obj = new SBI();
        obj.getInterestRate();
    }
}
```

A. 5.00% B. 6.00% C. Compilation Error D. Runtime Error

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46. What is the output of the below Java code?

```
Class CheckSring

{
    public static void main(String args[])
    {
        String s1 = new String(""nvipani"");
        String s2 = new String(""nvipani"");
        if (s1 == s2)
     System.out.println(""Equal"");

        else
          System.out.println(""Not equal"");
    }

}
```

A. Equal B. Not equal C. Compilation Error D. Runtime Error

47. Regular expressions are _____

A. Type 0 language B. Type 1 language
C. Type 2 language D. Type 3 language

48. The elements <DIV> and <SPAN> have the following characteristics

A. Element <DIV> inherits properties defined for <SPAN> in a stylesheet
B. Elements <SPAN> and <DIV> define content to be inline or block-level
C. <DIV> and <SPAN> are used as alternatives for the element <P>
D. <DIV> is used inside element <P>.

49. Given an arbitrary non-deterministic finite automaton (NFA) with N states, the maximum number of states in an equivalent minimized DFA is at least.

A. N^2 B) 2^N C. 2N D. N!

50. What does the following function does for a given binary tree?

```
int fun(struct node *root)
{
   if (root == NULL)
      return 0;
   if (root->left == NULL && root->right == NULL)
      return 0;
   return 1 + fun(root->left) + fun(root->right);
```

A. Counts leaf nodes
B. Counts internal nodes
C. Returns height where height is defined as the number of edges on the path from root to deepest node
D. Return diameter where diameter is the number of edges on the longest path between any two nodes

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51. Consider a situation where swap operation is very costly. Which of the following sorting algorithms should be preferred so that the number of swap operations are minimized in general?

A. Heap Sort B. Selection Sort C. Insertion Sort D. Merge Sort

52. Multithreading on a multi – CPU machine

A. decreases concurrency B.increases concurrency

C. doesn't affect the concurrency D.can increase or decrease the concurrency

53. Which of the following HTTP method should be used to fetch resources using RESTful web service?

A. GET B. SELECT C. FETCH D. POST

54. What is the return value of foo(6)?

```
int foo(int n)
{
    static int r = 0;
    if (n < =0)
    return 0;
    if (n% 2 = = 0)
    {
r = n;
return foo(n/2) + r;
    }
return foo(n-1) + r;
}
```

A. 8 B. 12 C. 9 D. 13

55. What is the output of the following code?

```
#include<stdio.h>
main()
{
    int n, i;
    n = foo(6);
    printf( "%d", n );
}
foo(int x)
{
     if(x == 2)
         return 2;
     else
     {
         printf("+");
         foo(x-1);
     }
}
```

A. ++++2 B. +++++2 C. +++++ D. 2

The End

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